# Assemble and disassemble

**Q-1** Carefully disassemble a desktop PC by removing each component (RAM, HDD/SSD, GPU, CPU, motherboard, PSU, etc.) in the correct order, and take a clear photo of each part as you remove it.

**ANS**: **Desktop PC Disassembly**

I carefully disassembled the desktop PC by following the correct order. Before starting, I shut down the computer, unplugged the power cable, and made sure the PC was completely disconnected from the power source.

| Step | Component Removed | Action | Photo |
|---|---|---|---|
| 1 | **Power Cable** | Turned off the PC and disconnected all power cables. | Paste Photo 1 |
| 2 | **GPU** | Removed the GPU power cable and unscrewed the GPU from the case. | Paste Photo 2 |
| 3 | **RAM** | Opened the RAM slot clips and carefully removed the RAM modules. | Paste Photo 3 |
| 4 | **Storage (HDD/SSD)** | Disconnected the SATA/data and power cables and removed the drive. | Paste Photo 4 |
| 5 | **CPU Cooler** | Disconnected the CPU fan cable and carefully removed the CPU cooler. | Paste Photo 5 |
| 6 | **CPU** | Opened the CPU socket lock and carefully removed the processor. | Paste Photo 6 |
| 7 | **Motherboard** | Disconnected remaining cables and removed the motherboard screws. | Paste Photo 7 |
| 8 | **Power Supply (PSU)** | Disconnected its cables and removed the PSU screws from the case. | Paste Photo 8 |

**Safety Precautions**

- The PC was completely powered off before disassembly.
- The power cable was removed before opening the case.
- Components were handled carefully by their edges.
- Screws and cables were kept safely so they could be identified later.
- Components were placed on a clean, dry, non-static surface.

**Q-2** Label each photo from your disassembly process with the component name and a short note about its connection type (e.g., SATA, PCIe, ATX power, DIMM slot).

**ANS: Disassembly Photo Labels**

| Photo | Component Name | Connection Type | Short Note |
|---|---|---|---|
| **Photo 1** | **GPU** | PCIe | The GPU is installed in the motherboard's PCIe slot and may also have a PCIe power cable. |
| **Photo 2** | **RAM** | DIMM Slot | The RAM module is installed in a DIMM slot on the motherboard. |
| **Photo 3** | **HDD/SSD** | SATA | The storage drive uses a SATA data cable and SATA power cable. |
| **Photo 4** | **CPU Cooler** | CPU Fan Header | The cooler fan connects to the motherboard's CPU_FAN header. |
| **Photo 5** | **CPU** | CPU Socket | The CPU is installed directly into the CPU socket on the motherboard. |
| **Photo 6** | **Motherboard** | ATX Power / CPU Power | The motherboard receives power through the 24-pin ATX connector and CPU power connector. |
| **Photo 7** | **PSU** | ATX Power | The PSU supplies power to the motherboard, CPU, GPU, storage, and other components. |

**Photo Label Format**

Under or beside each photo, write:

**Component:** GPU
**Connection:** PCIe

**Component:** RAM
**Connection:** DIMM Slot

**Component:** HDD/SSD
**Connection:** SATA

**Component:** PSU
**Connection:** ATX Power.

**Q-3** Safely store all removed components in anti-static bags or on an anti-static mat, and document your storage method with a photo.<br><br>Hint: Avoid placing components on metal or carpet surfaces to prevent static damage.

**ANS**: **Safe Storage of Removed PC Components**

After disassembling the desktop PC, I stored all the removed components safely to protect them from dust, scratches, and static electricity.

| Component | Storage Method |
|---|---|
| **RAM** | Placed inside an anti-static bag. |
| **GPU** | Placed inside an anti-static bag and kept on a safe surface. |
| **HDD/SSD** | Placed inside an anti-static bag. |
| **CPU** | Kept carefully in its protective packaging. |
| **Motherboard** | Placed on an anti-static mat/bag. |
| **PSU** | Kept on a clean, dry, non-metal surface. |
| **CPU Cooler** | Kept safely on a clean surface. |

**Safety Precautions**

- Components were handled by their edges.
- Components were **not placed on carpet or bare metal surfaces**.
- Anti-static bags or an anti-static mat were used for sensitive components.
- All screws and cables were kept separately so they could be identified during reassembly.

**Photo Evidence**

**Paste the photo here showing all removed components safely stored on the anti-static mat/bags.**

**Q-4**: Reassemble the PC step-by-step, applying thermal paste to the CPU before installing the cooler, and photograph each major step.

**ANS: Desktop PC Reassembly**

**After disassembling the PC, I carefully reassembled all the components in the correct order. Before installing the CPU cooler, I applied a small amount of thermal paste to the CPU.**

| Step | Reassembly Action | Photo |
|---|---|---|
| 1 | Installed the CPU into the motherboard CPU socket and locked it properly. | Paste Photo 1 |
| 2 | Applied a small amount of thermal paste on the top of the CPU. | Paste Photo 2 |

| Step | Reassembly Action | Photo |
|---|---|---|
| 3 | Installed the CPU cooler over the CPU and connected its fan cable to the CPU_FAN header. | Paste Photo 3 |
| 4 | Installed the RAM modules into the correct DIMM slots until the clips locked. | Paste Photo 4 |
| 5 | Installed the motherboard inside the PC case and tightened the screws. | Paste Photo 5 |
| 6 | Installed the HDD/SSD and connected the required SATA/data and power cables. | Paste Photo 6 |
| 7 | Installed the GPU into the PCIe slot and connected its power cable if required. | Paste Photo 7 |
| 8 | Installed the PSU and connected the motherboard, CPU, GPU, and storage power cables. | Paste Photo 8 |
| 9 | Checked all cables and connections and made sure no cable was blocking the fans. | Paste Photo 9 |
| 10 | Connected the power cable, switched on the PC, and checked that the system started normally. | Paste Photo 10 |

Thermal Paste Application

1. Make sure the CPU surface is clean.
2. Put a small amount of thermal paste in the center of the CPU.
3. Place the CPU cooler straight over the CPU.
4. Secure the cooler properly.
5. Connect the cooler fan cable to the CPU_FAN connector.

**Final Check**

After reassembly, I checked that the fans were working and the computer started normally. I also checked that the CPU and GPU were detected correctly.

**Photo Evidence**:
Paste the photographs of each major reassembly step above.

**Q-5** Power on the reassembled PC, enter the BIOS/UEFI setup, and take a screenshot or photo showing that the CPU, RAM, and storage devices are detected; then write a one-paragraph summary of any issues faced and how you solved them.

**ANS:**

**BIOS/UEFI Hardware Detection**

After reassembling the PC, I powered it on and entered the BIOS/UEFI setup by pressing the required key during startup. I checked the system information page and confirmed that the CPU, RAM, and storage device were detected correctly.

| **Component** | **Detection Status** |
|---|---|
| CPU | Detected |
| RAM | Detected – 16 GB |
| Storage | Detected – 1 TB SSD |

**Screenshot/Photo**:
Paste the BIOS/UEFI screenshot here showing the CPU, RAM, and storage information.

Issues Faced and Solution

After reassembly, the PC initially showed no display. I switched off the PC and checked the connections. I found that the RAM was not fully inserted into the DIMM slot, so I removed it and installed it again until the clips locked properly. After powering on the PC again, the display appeared and the BIOS/UEFI successfully detected the CPU, RAM, and SSD.